higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRONICS
NQF LEVEL 2

(10041012)

18 November 2019 (Y-Paper)
13:00–16:00

This question paper consists of 9 pages and 1 addendum.

Copyright reserved

TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each section on a NEW page.
5. Use only BLUE or BLACK ink.
6. Write neatly and legibly.

Copyright reserved

QUESTION 1

Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1–1.10) in the ANSWER BOOK.

1.1 Which ONE of the following factors does not influence the resistance of a material?

A Temperature
B Diameter
C Length
D Frequency

1.2 An insulator has …

A low resistance and allows current to flow.
B high resistance and does not allow current to flow.
C low resistance and does not allow current to flow.
D medium resistance and allows moderate current to flow.

1.3 If gold is the third band on a four-band resistor colour code, then the first two values are multiplied by …

A 100
B 1000
C 0.1
D 0.01

1.4 Electrical safety must be adhered to at all times in a working environment.

Which ONE of the following is NOT a cause of electric shock?

A Plugging two appliances into one plug
B Switching off a plug socket and then pulling an electrical cord out of it
C Faulty appliances
D Power lines that are down

1.5 According to …, current is directly proportional to voltage and inversely proportional to resistance.

A Lenz's law
B Faraday's law
C Ohm's law
D Kirchoff's current law

1.6 The following tools are required when soldering components onto a vero board or printed circuit board.

A Soldering iron, solder, pliers, desoldering pump or sucker, helping hands
B Soldering iron, solder, leather gloves, desoldering pump or sucker, helping hands
C Soldering iron, solder, pliers, desoldering pump or sucker, power supply
D None of the above

1.7 The joint in which the solder does not make good contact with the component lead:

A Warm joint
B Cold joint
C Tinning joint
D None of the above

1.8 The decimal equivalent of the binary number 1001001_2:

A 56_{10}
B 73_{10}
C 85_{10}
D 69_{10}

1.9 The output of a NOR gate is logic 1 if …

A all inputs are logic 1
B all inputs are logic 0
C one input is logic 1
D one input is logic 2

1.10 The electronic process that combines two or more lower bandwidth transmissions onto one higher transmission bandwidth:

A Impedance matching
B Modulation
C Multiplexing
D Network management

(10 × 1) **[10]**

Copyright reserved

QUESTION 2

2.1 What is the difference between a *permanent magnet* and an *electromagnet*? (2)

2.2 Explain the left-hand rule for a conductor. (3)

2.3 The resistance of a material depends on the following factors:

- Temperature
- Length
- Area
- Type of material

Explain the effect of area and temperature on resistance. (2)

2.4 Two resistors, 10 Ω and 20 Ω, are connected in parallel and this combination is connected in series with a 25 Ω resistor.

Calculate the total resistance of the circuit. (3)
[10]

QUESTION 3

3.1 What is meant by *forward bias*? (1)

3.2 What are free electrons according to the Electron Theory? (1)

3.3 What is the colour code for 1k2 ±10 % Ω? (4)

3.4 Explain how a P-type is formed. (2)

3.5 Explain the effect of temperature on a semiconductor material. (2)
[10]

QUESTION 4

4.1 Differentiate between a *dead* and a *live circuit*. (2)

4.2 Explain why earthing is important in electrical safety. (2)

4.3 Name any TWO causes of electric shock. (2)

4.4 Name any FOUR instances when emergency services need to be called if a person has experienced electric shock. (4)
[10]

Copyright reserved

QUESTION 5

5.1 Ohm's law formula is given as follows:

$$V = I \times R$$

Explain the effect of voltage and resistance on current. (2)

5.2 Voltage across a 10 Ω resistor is 15 V.

Calculate the current through this resistor. (2)

5.3 Use a circuit diagram to explain how to measure current through a load using a multimeter or ammeter. (2)

5.4 List FOUR consequences of not looking after instruments. (4)

[10]

QUESTION 6

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (6.1–6.10) in the ANSWER BOOK.

6.1 A side cutter can be used to remove a badly soldered component.

6.2 A good soldered joint should be brown and shiny.

6.3 A solder does not make good contact with the component in a cold joint.

6.4 Always work in a well-ventilated area when soldering.

6.5 The purpose of a wet sponge is to cool a soldering iron.

6.6 The advantage of a vero board over a breadboard is that no soldering of components is required.

6.7 All waste material can be disposed of in one container.

6.8 The lead at the bottom of the board can be bent at an 80-degree angle.

6.9 Glass cleaner can remove excess resin.

6.10 A dry joint can cause a poor electrical connection.

(10 × 1) **[10]**

Copyright reserved

QUESTION 7

7.1 What is the difference between a *breadboard* and a *vero board*? (2)

7.2 FIGURE 2 shows how to connect two resistors in series (see FIGURE 1) on a vero board planning sheet and FIGURE 4 shows how to connect two resistors in parallel (see FIGURE 3) on a planning sheet. Use the planning sheet on the ADDENDUM (attached) to show how you will connect the given circuit in FIGURE 5.

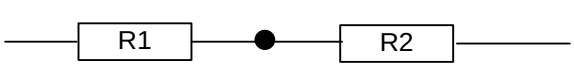


FIGURE 1
CIRCUIT DIAGRAM FOR TWO RESISTORS CONNECTED IN SERIES

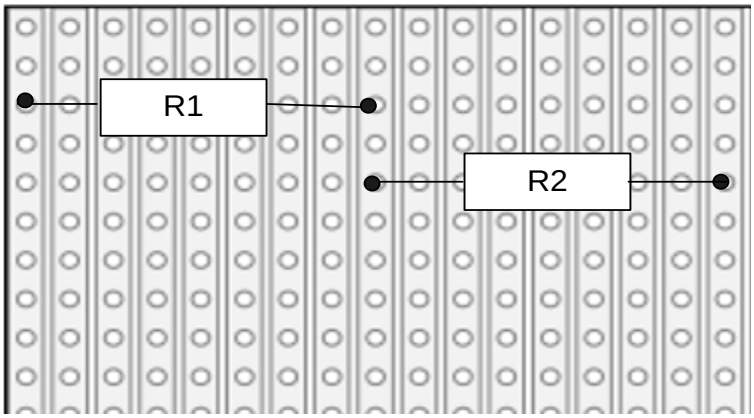


FIGURE 2
VERO BOARD LAYOUT FOR CIRCUIT IN FIGURE 1

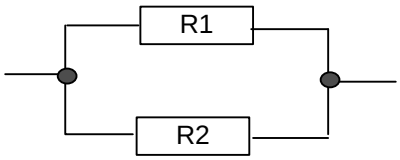


FIGURE 3
CIRCUIT FOR TWO RESISTORS CONNECTED IN PARALLEL

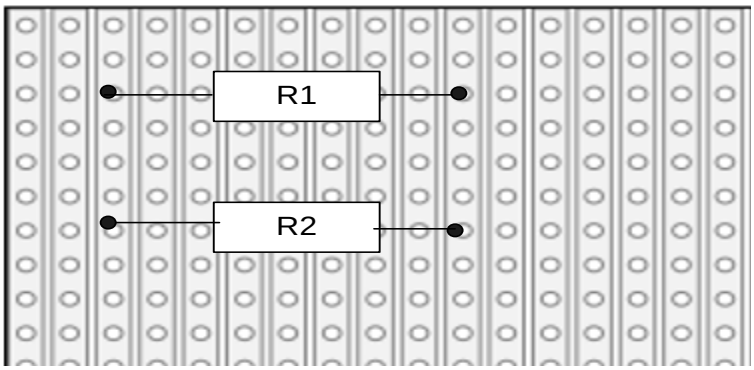


FIGURE 4
VERO BOARD LAYOUT FOR CIRCUIT IN FIGURE 3

Copyright reserved

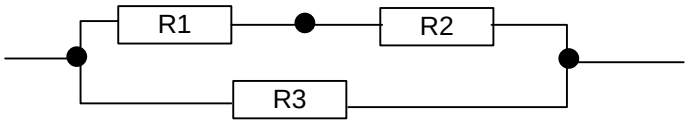


FIGURE 5
CIRCUIT DIAGRAM FOR RESISTORS CONNECTED IN SERIES-PARALLEL (3)

7.3 Electronics Level 2 students from the Thuto Ke Lesedi TVET College were asked to build a circuit using an NPN (BC108) transistor, 100 micro-Faraday (non-polarised) capacitor, two resistors and measure the current through R1 resistors as soon as a 12 V DC supply was connected.

Study FIGURE 6 below and name FOUR faults in the circuit.

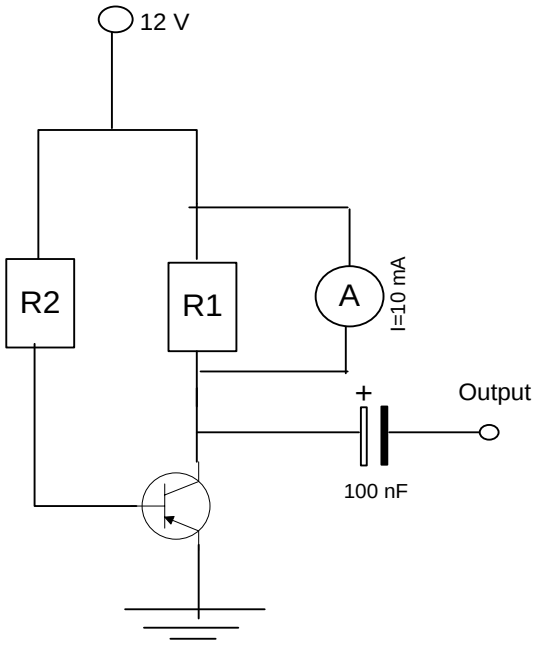


FIGURE 6 (4)

7.4 Name ONE instrument other than a voltmeter or multimeter to measure the output voltage in FIGURE 6. (1)
[10]

QUESTION 8

8.1 Convert 1001_2 to a decimal.

8.2 Calculate $1001_2 + 11001_2$.

8.3 Subtract 1001_2 from 11001_2.

8.4 Draw a switch circuit of a NOT gate.

8.5 Name the logic gate and draw its IEC symbol which gives an output of logic 1; only if one input is logic 1 or more inputs are logic 1.

(5 × 2) **[10]**

QUESTION 9

9.1 Name ONE discrete input and ONE analogue input that are applicable to PLCs. (2)

9.2 Describe how communication is established between a *programmer* and a *program.* (2)

9.3 One cannot merely plug any power supply into a PLC unit.

List FOUR steps you would take to insert a power supply into the PLC. (4)

9.4 Use symbols to explain what is meant by *normal closed input* and *de-energised output* in a PLC. (2)

[10]

QUESTION 10

10.1 Define *wavelength* according to concepts of telecommunication.

10.2 What is the difference between *amplitude modulation* and *frequency modulation*?

10.3 Explain the concept noise-to-signal ratio in relation to telecommunication media.

10.4 What does the abbreviations PSTN and PABX stand for?

10.5 Write only the abbreviation of a computer network that covers a …

10.5.1 wider area.

10.5.2 local area, such as an office, home or building.

(5 × 2) **[10]**

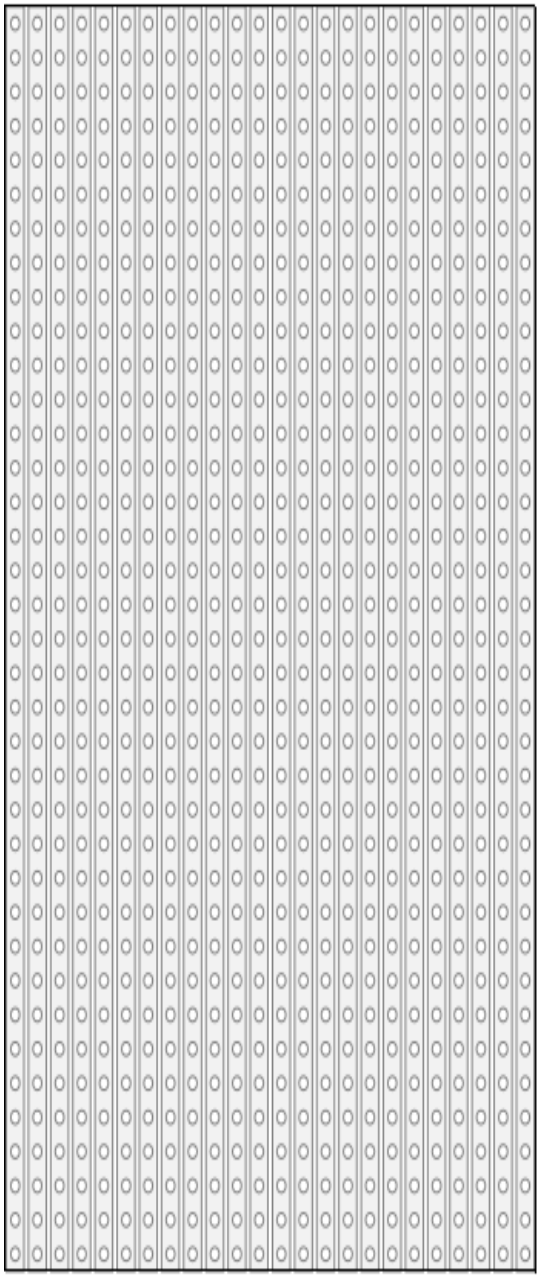
TOTAL: 100

Copyright reserved

(10041012)

ADDENDUM

EXAMINATION NUMBER:												
CENTRE NUMBER:												



Copyright reserved